

BERLIK, Kazakhstan

WMO: 353760

Lat: 49.883N

Lon: 69.517E

Elev: 1145

StdP: 14.10

Time Zone: 6.00 (E06)

Period: 94-08

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.0	-22.2	-31.8	1.0	-26.5	-27.2	1.3	-22.2	31.4	17.6	26.5	15.5	2.9	180	0.640

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.5	92.9	64.9	89.8	63.5	86.5	62.7	67.7	86.4	65.9	83.4	64.5	81.3	9.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.3	87.6	72.4	60.1	81.1	70.9	58.0	75.1	70.5	32.8	86.5	31.4	83.4	30.2	81.2	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.9	23.4	21.3	DB	-31.1	98.8	3.1	3.2	-33.3	101.1	-35.1	103.0	-36.8	104.8	-39.0	107.1	(4)
(5)			WB	-30.8	71.7	3.2	5.6	-33.1	75.7	-34.9	79.0	-36.7	82.1	-39.0	86.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.0	2.9	5.9	18.2	43.9	58.4	67.4	70.5	68.1	55.9	42.0	23.6	8.8		
	DBStd	26.95	14.50	12.19	14.07	11.64	8.35	7.03	6.01	7.78	8.69	9.50	13.29	12.47		
	HDD50	6345	1459	1235	986	246	26	1	0	0	42	280	792	1278		
	HDD65	10012	1924	1655	1450	635	230	53	19	58	291	712	1242	1743		
	CDD50	2324	0	0	1	62	288	524	634	563	219	33	0	0		
	CDD65	515	0	0	0	1	27	126	188	155	18	0	0	0		
	CDH74	7155	0	0	0	41	516	1880	2271	2047	382	18	0	0		
	CDH80	2815	0	0	0	7	136	768	903	875	125	1	0	0		
	WSAvg	8.1	7.7	9.9	8.6	9.0	8.5	8.2	7.8	7.1	7.0	7.1	7.9	8.3		
	PrecAvg	9.1	0.6	0.5	0.5	0.7	0.9	1.5	1.3	0.7	0.5	0.7	0.7	0.6		
(15) Precipitation	PrecMax	12.8	1.1	1.2	1.3	1.5	2.2	4.3	3.1	3.0	1.9	2.2	2.4	1.9		
	PrecMin	4.0	0.2	0.2	0.1	0.0	0.1	0.1	0.0	0.0	0.1	0.1	0.0	0.0		
	PrecStd	2.6	0.2	0.2	0.4	0.5	0.6	1.2	0.8	0.7	0.5	0.6	0.5	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.7	34.3	57.9	78.5	88.9	96.9	96.2	96.7	89.8	76.4	54.1	34.6		
		MCWB	32.6	32.7	47.5	58.0	62.1	64.5	68.5	65.2	60.6	53.5	45.9	33.0		
	2%	DB	31.1	31.7	47.3	72.7	83.9	91.5	91.8	92.5	83.1	69.7	48.1	31.7		
		MCWB	29.8	30.4	41.4	54.0	59.7	63.6	65.3	64.9	58.1	51.1	43.0	30.4		
	5%	DB	27.3	28.3	39.2	68.4	79.7	87.9	88.5	88.8	77.8	64.1	43.2	29.1		
		MCWB	25.9	27.0	35.8	51.8	58.1	62.2	64.4	63.3	56.2	48.4	39.5	27.6		
	10%	DB	24.1	24.5	35.1	63.2	75.7	84.3	85.0	84.7	73.2	58.3	39.2	26.8		
		MCWB	22.9	23.3	32.8	49.2	56.5	61.4	63.4	62.1	54.4	45.9	36.0	25.5		
	0.4%	WB	32.7	33.0	48.5	59.5	65.7	69.3	71.4	70.4	62.8	55.1	48.8	33.8		
		MCDB	33.3	34.0	56.1	75.7	82.8	89.0	88.7	88.6	87.4	70.5	52.5	34.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	29.9	30.7	41.7	55.5	62.0	65.9	68.2	66.5	59.8	52.5	43.2	30.5		
		MCDB	31.1	31.8	46.6	70.6	79.1	86.8	87.2	87.0	78.3	66.7	47.0	31.5		
	5%	WB	26.1	27.1	36.3	52.9	59.6	64.0	66.5	64.8	57.5	49.7	39.6	27.7		
		MCDB	27.2	28.3	39.2	66.1	75.3	83.2	82.1	84.4	74.8	61.5	43.1	28.9		
	10%	WB	23.0	23.4	33.1	50.0	57.6	62.3	65.1	63.2	55.1	47.0	36.2	25.5		
		MCDB	24.1	24.5	35.1	61.7	72.9	79.9	79.6	81.0	71.5	57.2	38.9	26.9		
	5% DB	MDBR	15.8	17.8	17.1	22.3	25.5	25.5	23.5	25.4	25.4	21.2	15.8	16.0		
		MCDBR	12.9	18.1	17.5	29.9	32.0	32.3	28.3	30.5	32.9	30.6	18.9	15.0		
		MCWBR	12.5	17.6	12.3	14.3	13.7	12.9	10.5	11.4	14.9	16.6	13.5	14.5		
		MCWBR	13.2	17.3	17.1	28.2	28.3	27.7	24.1	27.2	29.2	26.8	17.0	14.9		
(40) Clear-Sky Solar Irradiance	taub		0.296	0.330	0.381	0.412	0.412	0.411	0.428	0.402	0.354	0.342	0.328	0.298		
		taud	2.075	2.002	2.021	2.127	2.195	2.263	2.224	2.282	2.366	2.317	2.188	2.098		
	Ebn at Noon		225	246	255	263	268	269	262	264	266	246	212	204		
		Edh at Noon	27	37	44	45	44	41	42	38	31	27	24	23		
(44) All-Sky Solar Radiation	RadAvg		489	858	1261	1592	2004	2111	1965	1774	1345	761	375	331		
	RadStd		49	51	135	141	140	140	111	130	89	100	45	50		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page